# T2D Digital Twin

Forecasting hyperglycaemic events by fusing EHR and wearable data streams

**2 h**

early warning window

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Happiest Health Reimagining and Reforming Healthcare in India Summit 2026

# Diabetes care in India is still reactive

**101 M**

people living with diabetes in India (ICMR-INDIAB, 2023)

**Blind spots**

glucose spikes between clinic visits go unseen

**2 h**

ahead: the warning window this twin aims for

## Goal

A virtual replica of the patient that predicts adverse glycaemic events before they happen, and lets the doctor test lifestyle changes virtually.

# A digital twin fuses two data streams

## Static: EHR

- Age, sex, BMI
- HbA1c, fasting glucose, eGFR
- Diabetes duration, medication
- Genetic marker (TCF7L2)

**Twin model**

2-hour risk and what-if

## Dynamic: wearable

- Glucose (CGM), every 15 min
- Heart rate
- Steps
- Nightly sleep hours

*Focus condition: Type 2 Diabetes*

# Privacy-safe synthetic data

**300**
virtual patients

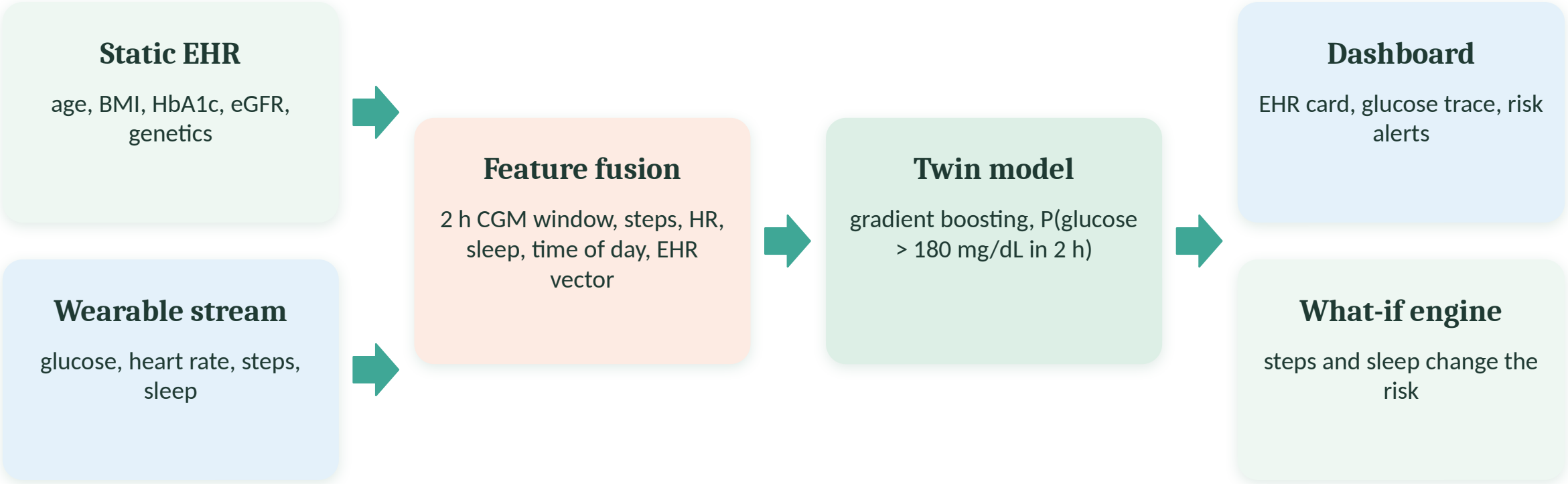
**14 days**
of monitoring each

**15 min**
sampling resolution

## How the simulator works

- Meal-driven glucose spikes from Indian-diet carbohydrate loads
- Scaled by insulin resistance from BMI, HbA1c, TCF7L2 alleles and metformin
- Post-meal walking lowers the spike; short sleep raises it
- No real patient data: DPDP Act and HIPAA safe

# System architecture




*Evaluation: ablation of CGM only vs + wearable vs + EHR, with ROC-AUC, PR-AUC and permutation importance*

# How the twin learns

**8**

CGM window features: level, spread, trend over 2 h

**7**

wearable and time features: steps, heart rate, sleep, hour

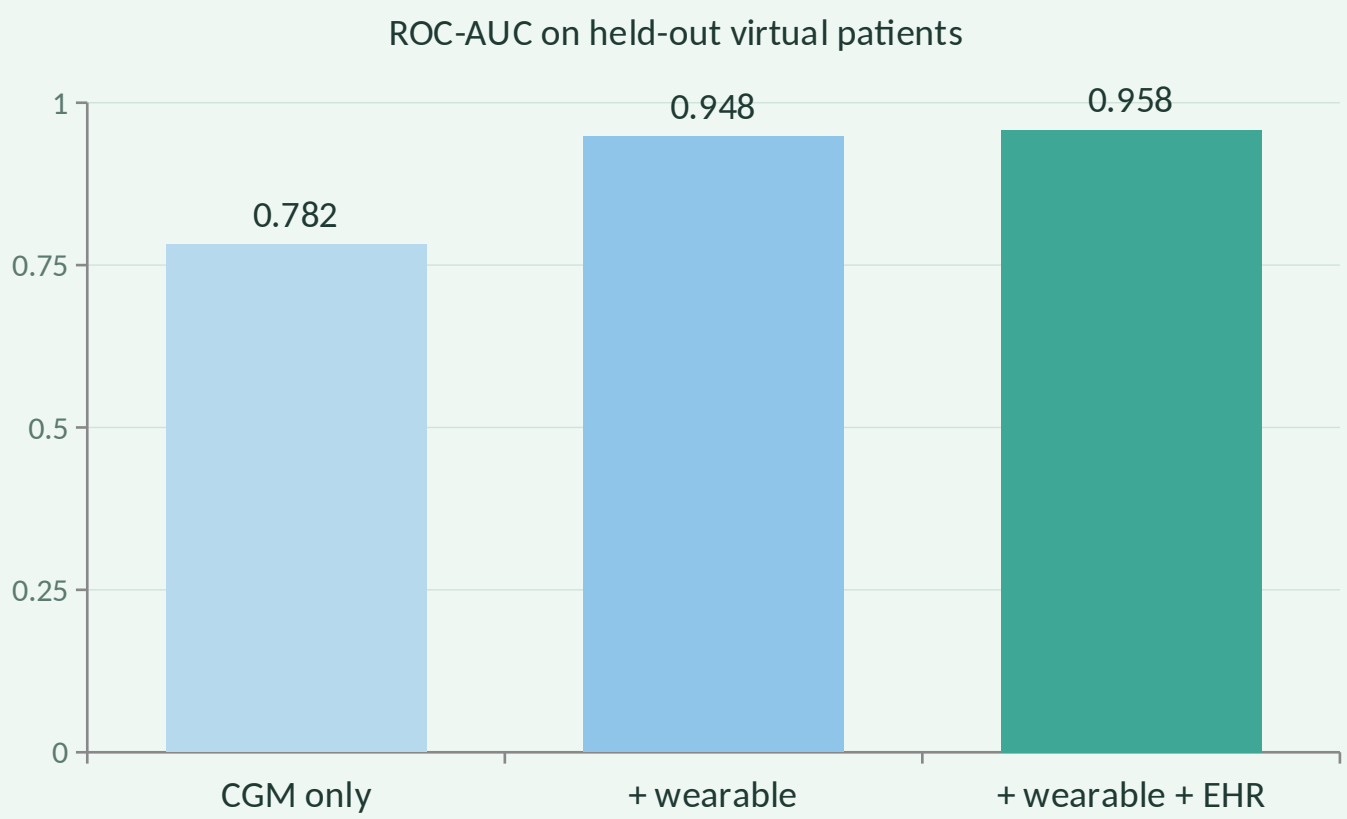
**11**

EHR features: demographics, labs, genetics, medication

## Target and training

Will glucose exceed 180 mg/dL within the next 2 hours, given it is at or below 180 now? Gradient boosting classifier, trained on 75% of patients and tested on unseen 25% (no leakage).

# Fusing both streams wins




**0.958**

ROC-AUC, full twin

**0.79**

precision

**0.71**

recall

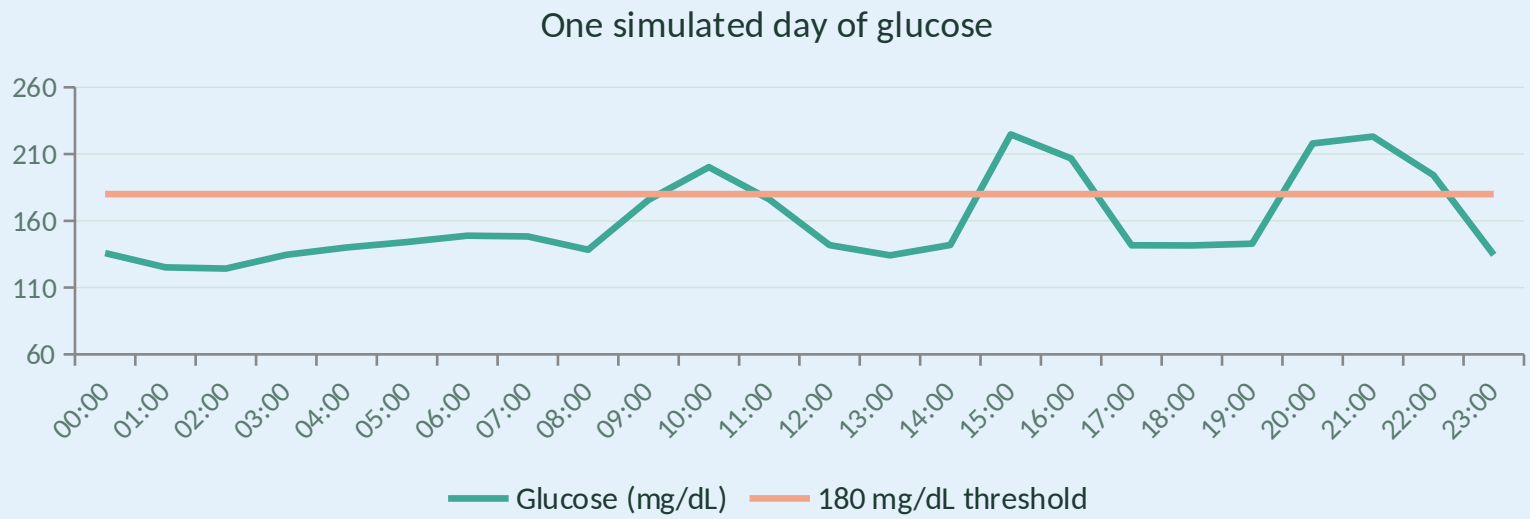
*Synthetic data: an upper bound, not clinical evidence*

# Doctor's view of the virtual patient

## Patient #3

EHR card: age, BMI, HbA1c, eGFR, medication, genetic risk, with live glucose, sleep and steps

**Alert when 2 h risk passes the chosen threshold**




## What-if simulation

Slide steps or sleep up or down and the twin re-scores the 2 hour risk instantly.

# Why it matters, and what to be honest about

## Relevance to India

- Huge and growing T2D burden
- Carbohydrate-heavy diets
- Low-cost wearables and CGM are spreading
- Proactive alerts support earlier intervention

## Limitations

- Synthetic data: results are an upper bound
- Simulator encodes assumed physiology
- Predicts high glucose only, not lows
- Not a medical device; not clinically validated

# Next steps

1

## Validate on real data

Open, de-identified CGM datasets and Synthea / MIMIC-IV EHR

2

## Richer signals

HRV, sleep stages, meal logging, Indian food database

3

## Better models

LSTM / TCN sequence models, SHAP explanations

4

## Clinician pilot

Test usefulness and alert fatigue with doctors

## Thank you